



جامعة الكفيل
University Of Alkafeel



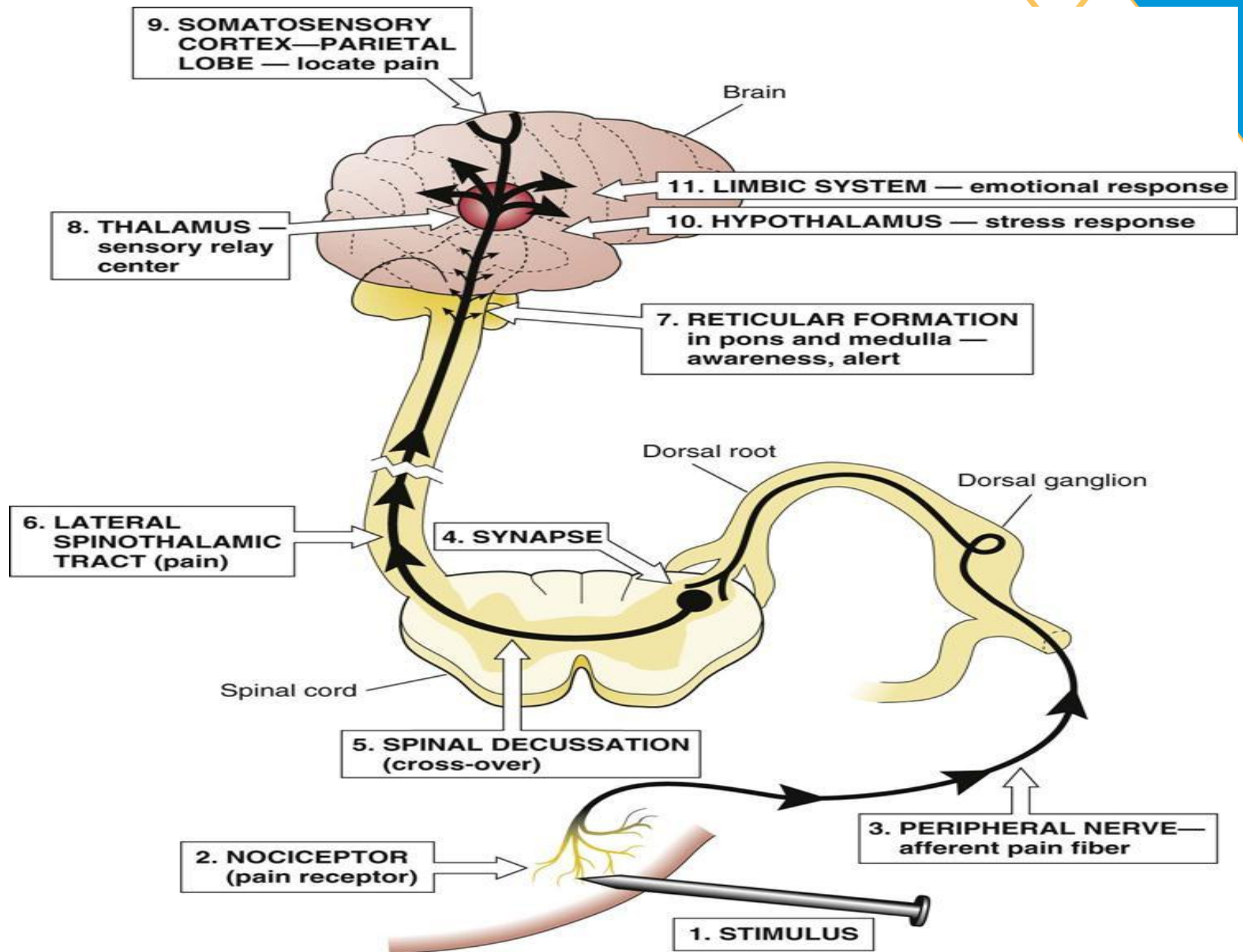
Narcotics analgesics



Pharma for dentists

- **Algesia (pain)** an ill defined as, unpleasant sensation, usually evoked by an external or internal noxious stimuli.
- Its a warning signal and primarily protective in nature, but causes discomfort.
- Excessive pain may be unbearable, anxious, causing nausea, sweating, palpitation, rise or decrease B.P.





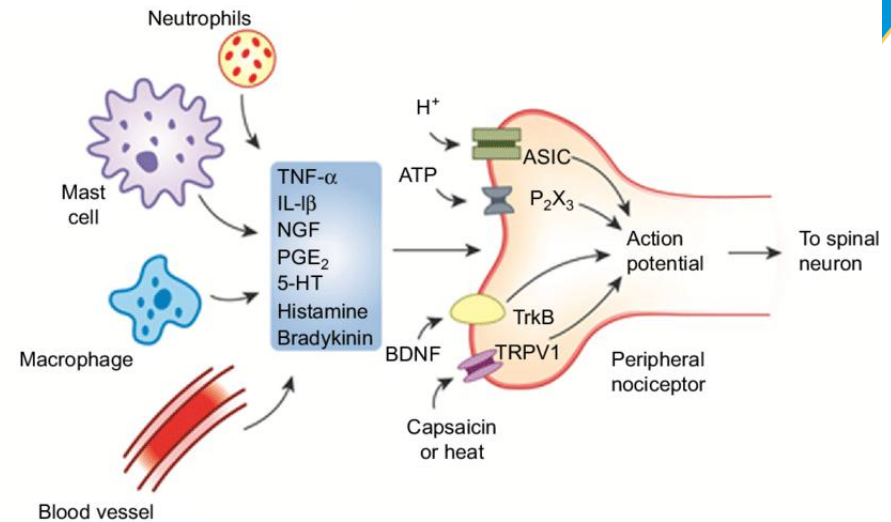
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- **Analgesics** are drugs that selectively relieve pain by acting on CNS or peripheral pain mechanisms.



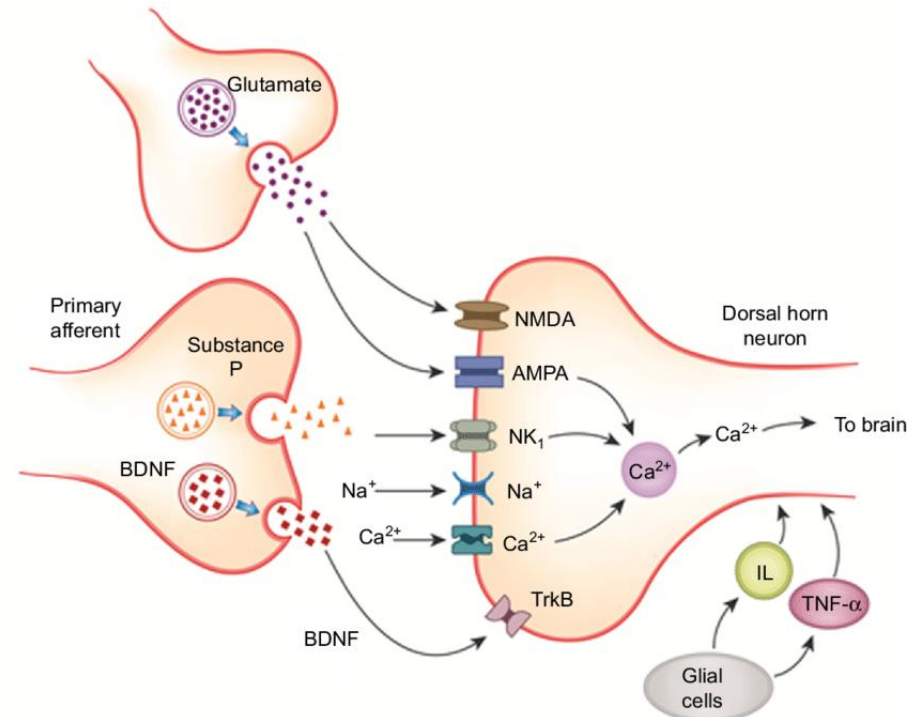
Pain Perception

- Activation of pain receptors (nociceptors) in peripheral tissue (muscle, skin, deep viscera) by a number of chemical mediators (histamine, bradykinin, PG, ATP & lactic acid) that released at the site of tissue damage.
- Transfer of pain information from nociceptors in the periphery to the spinal cord. Excitatory A.A (glutamate, neurokinins "sub. P & neurokinin A") mediate such synapse connection.
- Onwards passage of pain information to higher centers.

A Peripheral sensitization



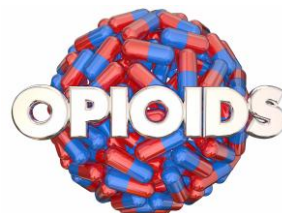
B Central sensitization



► Analgesic drugs divided into:

A. Narcotic/Opioids/Morphine-like

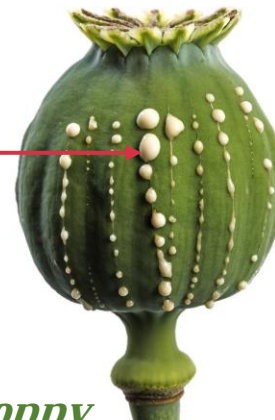
B. Non-narcotic/NSAIDs/Aspirin-like



- **Opioids** are natural or synthetic compounds that produce *morphine-like effects*.
- **Opiates** is reserved for drugs, such as *morphine and codeine*, obtained from the juice of the **opium poppy**.
- Although the opioids have a broad range of effects, their primary use is to relieve intense pain and the anxiety that accompanies it, whether that pain is from surgery or a result of injury or disease, such as cancer.



morphine



opium poppy



Opioids:

- Natural or synthetic compounds that produce morphine-like effects.
- Act by binding to specific receptors in the CNS to produce effects that mimic the action of endogenous peptide neurotransmitter **(e.g. leu- and met-enkephalins)**
- Although the opioids have a broad range of effects, their primary use is to relieve intense pain and the anxiety that accompanied it



Opioids receptors:

- 4 families of receptors which are: mu, kappa, delta, and sigma. Each of these receptors exhibit different specificity for the drug that they bind.

- ❖ μ (mu)
- ❖ κ (kappa)
- ❖ δ (delta)
- ❖ θ (sigma)





μ -Receptors are thought to be responsible for most of the analgesic effects of opioids, and for some major unwanted effects (e.g., respiratory depression, euphoria, sedation and dependence).

Most of the analgesic opioids are **μ -receptor agonists**.

δ -Receptors are probably more important in the periphery but may also contribute to analgesia.

κ -Receptors contribute to analgesia at the spinal level and may elicit sedation and dysphoria, but produce relatively few unwanted effects and do not contribute to dependence. Some analgesics are relatively κ -selective.

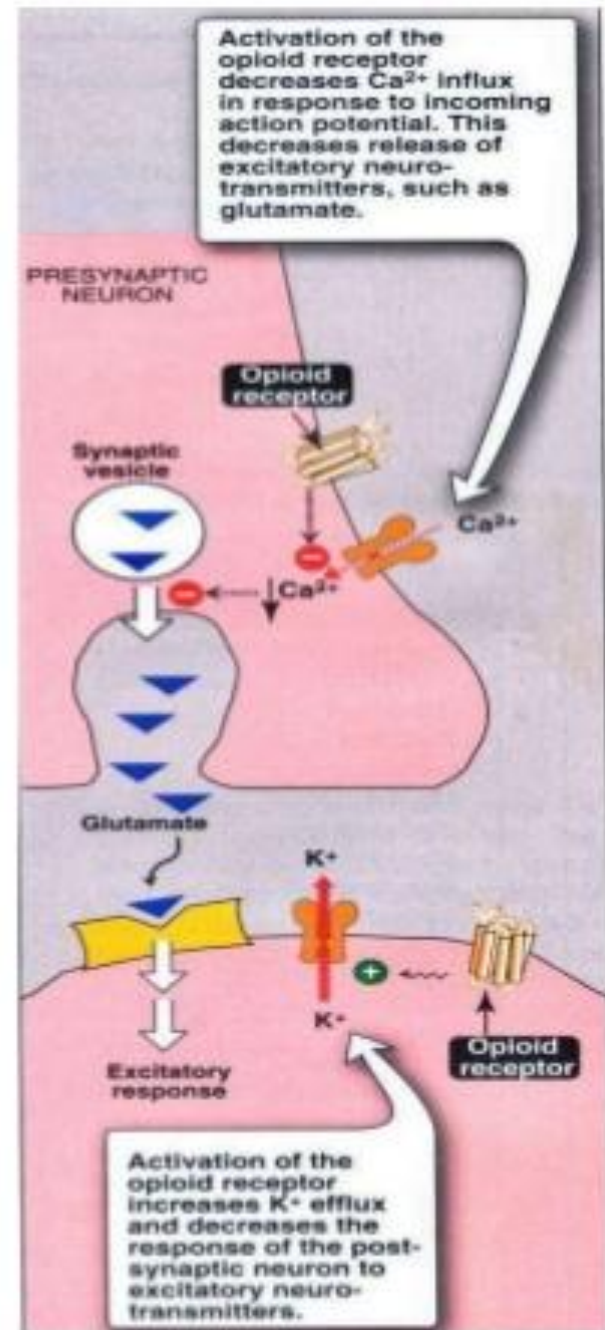
Mechanism of Action

All opioid receptors are G-protein coupled receptors and inhibit adenylate cyclase.

They are also involved in

- Postsynaptic hyperpolarization (increasing K^+ efflux)
- Reducing presynaptic Ca^{++} influx

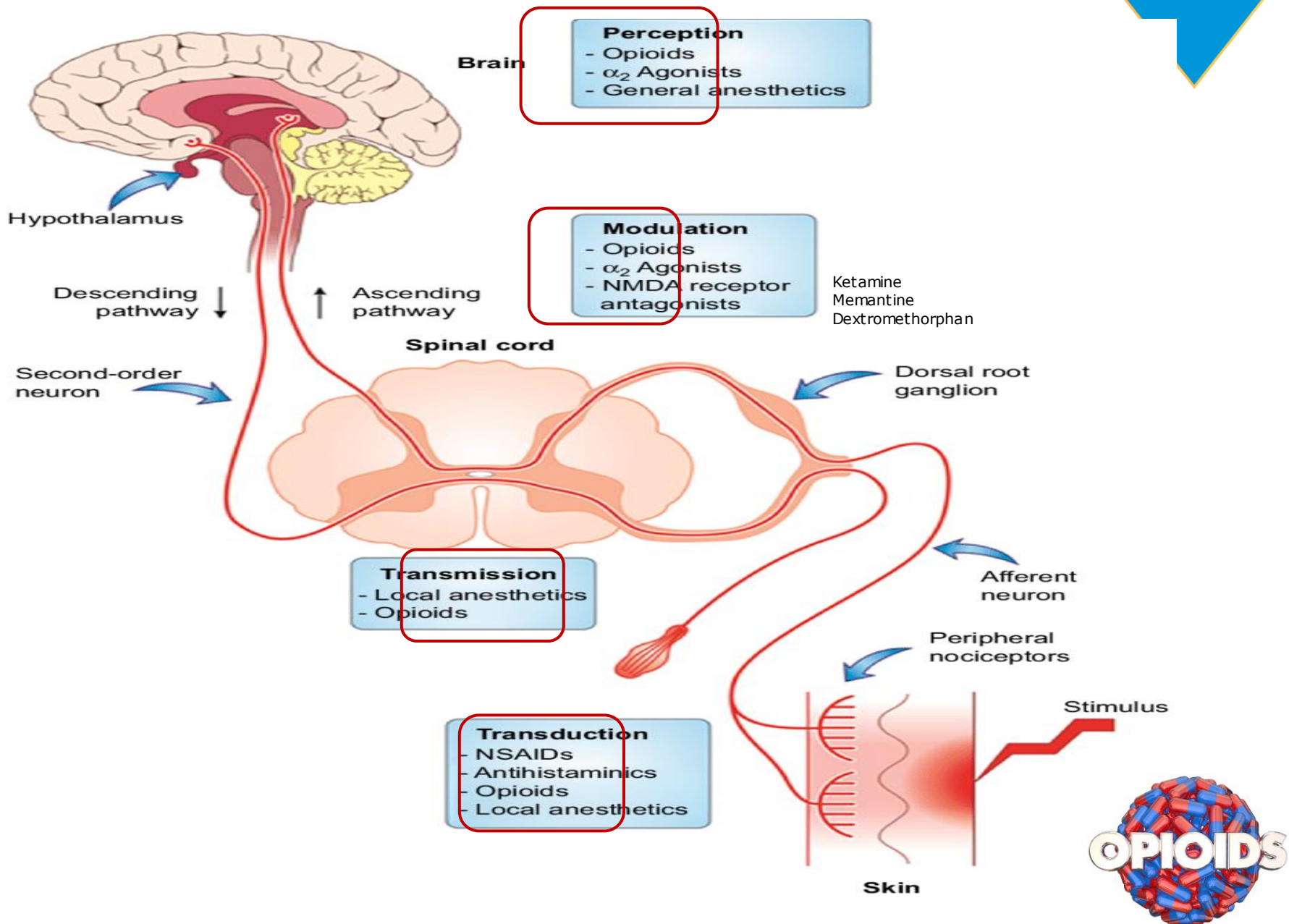
thus inhibits neuronal activity.



Opioids peptides:

- -Opioid receptors are activated by endogenous peptides under physiological conditions.
- -These peptides (**e.g. enkephalins , dynorphin, β -endorphin**) are synthesized in cell body and transported to nerve ending
- -On release from nerve endings, they bind to opioid receptors and can be displaced from binding by opioid antagonists.
- -Opioid peptides appear to modulate transmission at many sites in the brain and spinal cord and in peripheral sensory nerves.
- -Opioid peptides are also found in the adrenal medulla and neural plexus of the gut.

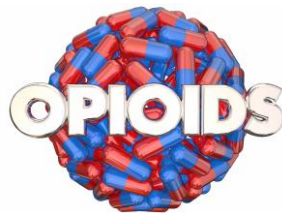




Classification of opioids:

A-Spectrum of clinical uses:

- **Analgesics** e.g., morphine , fentanyl
- **Antitussives** e.g., codeine , dextromethorphan
- **Antidiarrheal** e.g. diphenoxylate ,
loperamide



► **Classification of opioids:**

B-Strength of analgesia:-



B-Strength of analgesia:-

a. Strong agonists:

- Morphine
- Meperidine (Pethidine)
- Methadone
- Fentanyl
- Heroin (Diamorphine)

b. Moderate agonists:

- Codeine
- Dextromethorphan
- Diphenoxylate
- Loperamide
- Dextropropoxyphene

c. Partial agonist:

- Pentazocine
- Phenazocine

d. Antagonists:

- Naloxone
- Naltrexone
- Nalorphine
- Nalmefene

Must important classification



Classification of opioids:

According to the Strength of analgesia

1-STRONG AGONISTS:

- **Morphine** is the prototype strong agonist
- Other **strong agonists** include: meperidine, fentanyl, diamorphine (heroin), methadone.
- -These drugs show high affinity for **μ receptors** and varying affinities for κ and δ receptors

Actions of morphine:

a-Analgesia

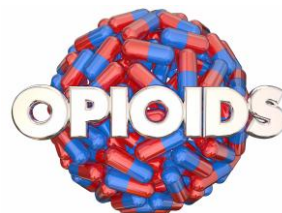
- Morphine causes analgesia (relief of pain without loss of consciousness)
- Opioids relieve pain both by raising the pain threshold at the spinal cord level and, more importantly, by altering the brain's perception of pain.



Actions of morphine:

b-Sedation and euphoria:-

- These effects may occur at doses below those required for maximal analgesia.
- The sedation is additive with other CNS depressants, but there is little amnesia.
- Some patients show dysphoric effects from opioid drugs.



Actions of morphine:

c-Respiratory depression:

- Morphine causes respiratory depression by reduction the sensitivity of respiratory center neurons to CO₂
- At higher doses, there will be respiratory arrest.
- Respiratory depression is the most common cause of death in acute opioids overdose.



Actions of morphine:

d-Depression of cough reflex:-

- Both morphine and codeine have antitussive properties.
- Cough suppression does not correlate closely with analgesic and respiratory depressant properties of opioid drugs.
- The receptors involved in the antitussive action appear to be different from those involved in analgesia.



Actions of morphine:

e-Miosis:

- Pin point pupil is characteristic of morphine poisoning which occurs as a result of stimulation of mu and kappa receptors.
- Morphine stimulates 3rd nerve nucleus which cause enhancement of the parasympathetic stimulation to the eye and thus miosis.
- There is only little tolerance to this effect (even addicts show pin point pupil).
- This is an important diagnostic feature of coma because most other causes of coma cause dilation of the pupil.



Actions of morphine:

f-Emesis:

- Morphine directly stimulates the chemoreceptor trigger zone that causes emesis.



g-GIT:

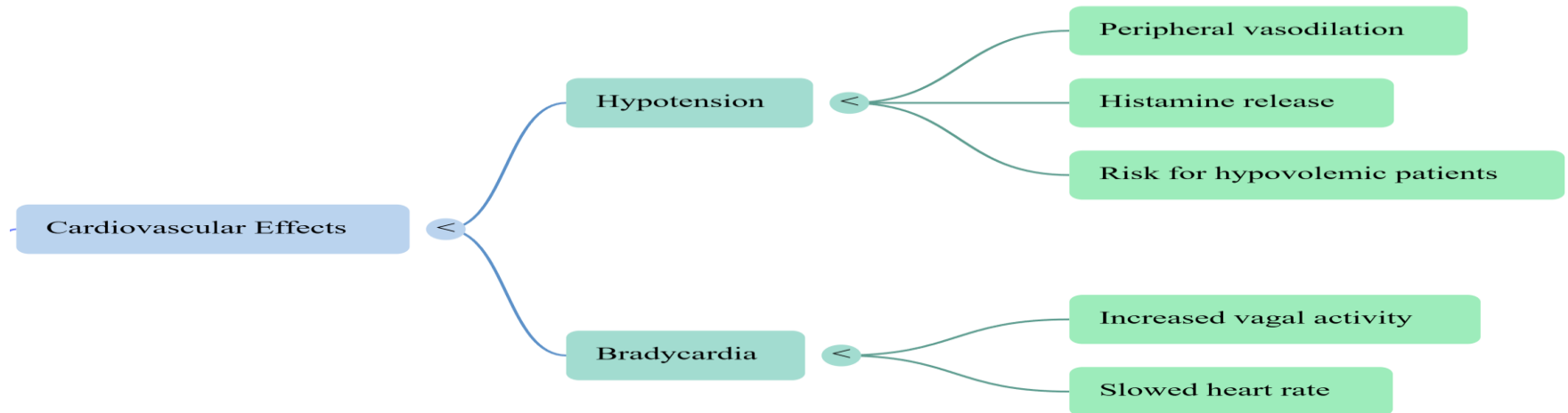
- Morphine relieves diarrhea by decreasing the motility and increasing the tone of intestinal smooth muscle.



Actions of morphine:

h-Cardiovascular:

- In large doses, morphine may cause hypotension and bradycardia



- Morphine is contraindicated in severe brain injury. **why??**



Actions of morphine:

i-Histamine release:

- Morphine is contraindicated in patient with asthma. **why ??**
- Morphine may cause urticaria, sweating and vasodilation. **why?**



Actions of morphine:

j-Hormonal actions:

- Inhibit release of gonadotropin-releasing hormone
- Inhibit release of corticotrophin-releasing hormone
- Decrease the concentration of:
 - luteinizing hormone
 - FSH
 - ACTH
 - and endorphin β
- Testosterone and cortisol level decrease
- Increase prolactin and GH release by diminishing dopaminergic inhibition



Actions of morphine:

k-Chronic effects:

A-Tolerance:

-Marked tolerance can developed to acute pharmacological effects, with exception of miosis and constipation.

B-Dependence:

-Is a physiological response to chronic therapy with opioid especially strong agonists.

-Abstinence syndrome (withdrawal syndrome) : **abrupt discontinuance of opioid may cause rhinorrhoea, lacrimation, chills, muscle ache, diarrhea, anxiety and hostility**

► Therapeutic uses of morphine

1-Analgesia:

- Treatment of constant moderate to severe pain.

2-Acute pulmonary edema:

- I.V morphine dramatically relieves dyspnea caused by pulmonary edema associated with LVF –possibly by vasodilator effects.

3-Anesthesia:

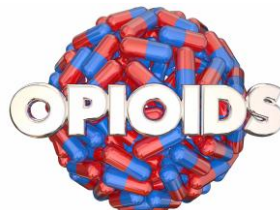
- Opioids are used as preoperative medication and as intraoperative agents.

- High –dose I.V morphine and fentanyl are the major components of anesthesia of cardiac surgery



Side effects

- **M**iosis
- **O**rthostatic hypotension
- **R**espiratory depression
- **P**ain suppression/ **P**ruritus
- **H**istamine release/**h**ormonal release
- **I**ncreased intracranial tension.
- **N**ausea
- **E**uphoria
- **S**edation



Fentanyl:

- Administered **parenterally**
- It has **shorter duration** of action(1-2 h) than **morphine**
- Available as **transdermal patch** which provide analgesia for 72 hrs



■ Meperidine (pethidine):

- It has **shorter duration** of action (2-3 h) than morphine



Methadone:

- -Act mainly at μ receptors
- Longer duration of action than morphine (24 h)
and this allow using methadone in:-
 - 1-Chronic pain (palliative care)
 - 2-treatment of opioid withdrawal



■ Diamorphine (Heroin):

- **The most potent** of all dependence-producing opioids.
- **More lipid-soluble** than morphine.
- **Rapid onset of action** (than morphine)
- Produce greater euphoria than morphine.
- **Not approved** for clinical use.



Thank you
for Listening

